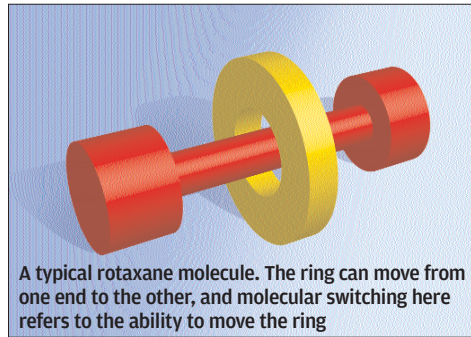


Molecular Switches

Since we're talking miniature, let's go all the way: how about storing a bit in a single molecule? Short of using a single atom—which is more difficult—this is about as small as it gets. (Later, we do talk, though, about single atoms!) Like we've said, it's all about 1s and 0s; so what one needs is a molecule with two stable states, and a means of switching the molecule between these states.

How Does It Work?

A molecule currently being researched is "rotaxane." Now this may sound way out: the rotaxane molecule has a thread-like section, with a ring structure that moves around it. The ring doesn't fall off because of the dumbbell-type bulbs at the ends of the thread. Ways have been found to move the ring from one end of the thread to the other, both of which are stable states for the molecule.



The reading procedure is complex, and cannot be discussed in this space.

What Is Being Done?

Scientists successfully demonstrated, in January 2003, that they could store information using rotaxane molecules. They reportedly achieved storage densities in the range of 1 to 10 gigabits per square inch.

What Promise Does It Hold?

There's currently more research going on in terms of using molecular switches as logic gates and such. But researchers are already envisaging fully-molecular computers, where the storage mechanisms, too, would be molecular. This is looking way into the future, but fully-molecular computers are an important idea—they represent the smallest you can get!

Atomic Storage Using STMs And AFMs

Take a look at the figure alongside. Notice the "IBM"? It's not a photograph, but it depicts what researchers have etched at the atomic scale: each of the little hills is an *individual xenon atom*! The thing was produced using an Scanning Tunnelling Microscope (STM) operating a few degrees above absolute zero.

How It Works

An STM has the ability to give a view of surfaces at the atomic scale, and researchers have envisioned the application of the technique to achieve ultra-high-density storage. The STM has an ultra-sharp tip placed extremely close to the substrate being written onto. A voltage applied between the tip and the substrate gives rise to a tunnelling current. The tunnel current depends on the separation between the tip and the substrate. As the tip is moved over the surface, the tunnel current is monitored, and the position of the tip is changed such that the current is constant—this way, the topology of the surface can be mapped out. The beauty of the STM is that it can be used not only to map a surface, but also to modify it.

There are difficulties with the STM approach—one is the problem of maintaining the distance between the tip and the surface at the angstrom level (an angstrom is 0.1 nm). To

overcome these difficulties, researchers are concentrating more on the Atomic Force Microscope (AFM). Here, the tip rests on a cantilever spring. This allows for two things: first, the tip can actually touch the surface, because of the "bounce" enabled by the spring. Second, by monitoring and controlling the spring, extremely small forces can be sensed as well as applied.

What's Being Done

The letters "IBM" have been etched on a surface using an STM, as mentioned above. Researchers are playing around with the idea of using an AFM in contact with a hard disk-like surface to etch data pits. It will take a long time for this to materialise, but remember that we're talking about a "hard disk" that writes *individual atoms*! Disk storage just cannot get any denser than that now, can it?

What Promise Does It Hold?

The use of STMs, AFMs, and similar devices are almost the ultimate application of nanotechnology to data storage. The potential storage capacities are enormous—for example, the etched letters in the figure represent a storage density of 1 *million* gigabits per square inch! That's 2 lakh CDs on a square inch!

